

KalaSetu AI (कलासेतु)

Next-Generation Rural Artisan Market Linkage, Conversational Cataloging, Verifiable Provenance, and Conscious Heritage Commerce Ecosystem.

✦ **Core Ecosystem** Discover → Understand the Story → Experience the Craft → Support the
Direction: Artisan → Purchase → Continue Supporting

TARGET COMMUNITY

**7M+ Indian Rural
Artisans**

PRIMARY INNOVATION

**Voice-to-Catalog &
Provenance**

AI RUNTIME COST

**100% Free / Open
Source**

DELIVERY MODALITY

**Android APK & Offline
PWA**



Section 1: The Critical Problem & Existing Market Loopholes

India's traditional handicrafts and handlooms represent the country's second-largest employment sector after agriculture. Despite immense global demand, the rural artisans who preserve this heritage face systemic exclusion from the digital economy.

1. The "Retailer-Connection" Trap

Existing e-commerce giants (Amazon Karigar, Etsy, Flipkart Samarth) primarily connect buyers with urban retailers and aggregators. The rural creator remains anonymous, invisible, and dependent on exploitative middlemen who take 60%–70% margins.

KalaSetu's Direct Artisan Attribution

Zero-middleman direct linkage. Every product features an immutable digital **Artisan Pehchan Card**, direct QR contact, and personal audio lineage. Buyers connect directly with the maker.

2. The Digital Literacy & Form Barrier

Conventional seller portals require typing, complex SKU categorization, GST details, English descriptions, and high-spec desktop setups—impossible for rural artisans.

Voice-First Vernacular Cataloging

The artisan speaks in their local dialect for 10 seconds: "बगरू हैंड-ब्लॉक दुपट्टा, प्राकृतिक नील रंग, 2 दिन लगे, 1200 रुपये". KalaSetu's AI extracts, structures, and creates luxury listings instantly.

3. Counterfeits & Unverifiable Claims

Industrial powerlooms and factory machine-prints sell cheap polyester as "Pure Handwoven Silk" or "Natural Indigo", destroying the market value of authentic handmade crafts.

ClaimSafe AI & Verifiable Product Passport

AI scans 16+ high-risk claims, enforces hard publishing gates, and issues a tamper-evident Product Passport verifying craft technique, materials, and GI heritage cluster.

4. The Financial Blindspot (Unclaimed Schemes)

Over ₹13,000 Cr in government support (PM Vishwakarma, Pehchan IDs, Mudra loans) remains unclaimed by rural artisans due to lack of awareness and complex paperwork.

Kala Funds Financial Aid Hub

An integrated scheme engine where artisans discover eligible welfare programs, toolkits, collateral-free credit, and NGO grants in 1 tap directly from their phone.



Section 2: Complete KalaSetu AI Master Feature Catalog

16 Verified Features



1. Voice-to-Catalog Smart AI

Multilingual speech recognition (14+ Indian dialects). Converts casual voice notes into structured product drafts with title, materials, dimensions, and craft classification.



2. ClaimSafe AI Verification

Scans 16+ sensitive claims (*100% pure silk, GI Tagged, Natural dye*). Hard publishing validation gate and public text sanitizer to prevent buyer misrepresentation.



3. FairPrice Guard & Wage Defense

Cost-plus formula calculator enforcing fair artisan hourly rates: $\text{Cost} = \text{Raw Materials} + (\text{Hours} \times \text{Rate}) + \text{Overhead}$. Warns against ₹0 labor and predatory middleman discounts.



4. Verifiable Product Passport

QR-code enabled digital provenance certificate. Displays verified material claims, GI cluster attribution, artisan lineage story, and strict privacy consent defaults.



5. Truthful Image Studio

Safe image enhancement policy (strictly non-synthetic, preserves authentic handmade imperfections). Live before/after toggle and real-time blur, underexposure, and glare diagnostics.



6. Craft-Aware Missing Field Wizard

Contextual templates for Bagru Block Prints, Jaipur Blue Pottery, Bastar Dhokra, etc. Non-intrusive step-by-step wizard prompting for vital buyer specifications.



7. Offline-First Drafts & Conflict Sync

IndexedDB v2 local storage with retry queue. Operates 100% offline in rural villages. Timestamp conflict management creates safe copies without silent overwrites.



8. Facilitator & NGO Review Desk

5-metric audit dashboard for cluster coordinators and SHG leaders. Queue for inspecting pending claims, vetting fair prices, and submitting audit field notes.



9. Cluster Geo-Fencing Dispatch

Validates that product dispatch origin physically matches the recognized geographical heritage boundary (e.g. Bagru cluster polygon), proving authentic provenance.



10. Kala Reels → Direct 1-Tap Buy

Short-form making videos showcasing raw artisanal process. Floating buy-now button allows consumers to discover, learn, and purchase directly inside the video flow.



11. AI Room Décor Simulator

Space Stager: Buyers upload a photo of their living room, empty wall, or corner table. AI visualizes how authentic vases, tapestries, or brass lamps fit their home.



12. AI Multi-Object Visual Search

Uploads a lifestyle or Pinterest image and detects multiple objects simultaneously (e.g., [Terracotta Vase] + [Jute Dhurrie]), matching each to KalaSetu catalog items.



13. Kala Funds (Financial Aid Hub)

Direct discovery of government schemes: PM Vishwakarma (toolkit grants & ₹3L collateral-free loans), AHVY, Mudra loans, and Pehchan artisan card benefits.



14. Artisan Productivity Score

Business health score (0-100) based on fulfilled orders, catalog updates, hisab ledger consistency, and buyer ratings. Enhances loan creditworthiness.



15. KalaSetu Saathi (Voice AI Companion)

Conversational agent assisting buyers in finding heritage gifts and assisting artisans with profit calculations and daily stock status in regional languages.



16. Patron & Artisan Reward System

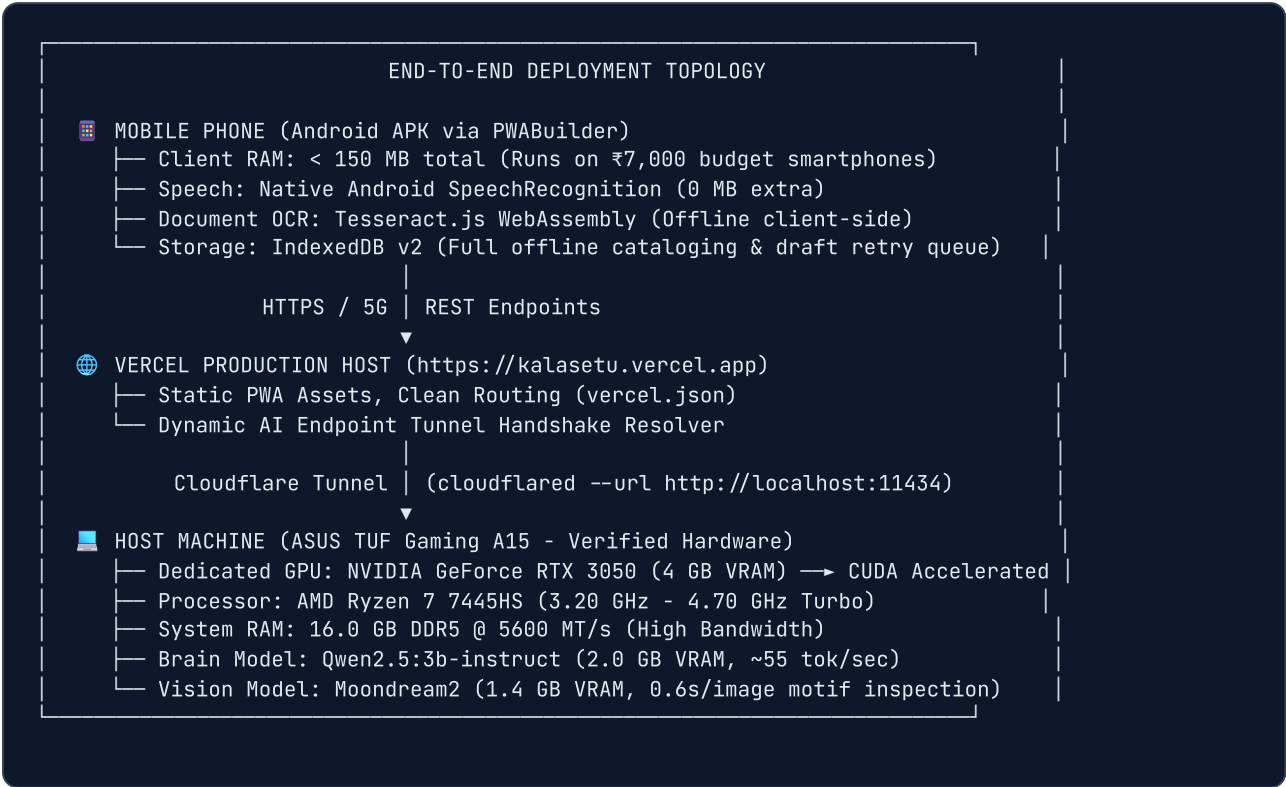
Artisan Karma points for verified sustainable dyes; Heritage Patron badges for repeat buyers supporting rural craft clusters.



Section 3: Technical System Architecture & Hardware Footprint

Zero Cloud Billing

KalaSetu AI is engineered for extreme resource efficiency. The architecture decouples the lightweight mobile presentation layer from the high-throughput local AI inference engine, enabling native mobile APK delivery while running cutting-edge models without cloud subscriptions.



SYSTEM LAYER	TECHNOLOGY	HARDWARE ROLE	MEMORY FOOTPRINT
Mobile Client	Android TWA / PWA (PWABuilder)	Artisan UI, Mic, Camera, Offline DB	< 150 MB RAM
Language Model	Qwen2.5-3B-Instruct (Q4_K_M)	Catalog structuring & negotiation reasoning	2.0 GB VRAM (RTX 3050)
Vision Model	Moondream2 (1.86B Q4)	Motif identification & craft photo inspection	1.4 GB VRAM (RTX 3050)
Document OCR	Tesseract.js (v5) + Devanagari	Handwritten expense slips & Pehchan ID OCR	80 MB Client WASM
Cloud Bridge	Cloudflare Tunnel (cloudflared)	Zero-config HTTPS tunnel to mobile APK	15 MB System RAM



Section 4: Kala Funds — Government Schemes & Financial Aid Hub

Financial Inclusion

SCHEME / PROGRAM	TARGET BENEFICIARY	KEY FINANCIAL / MATERIAL BENEFIT	KALASETU INTEGRATION
PM Vishwakarma Yojana	18 traditional artisan trades (Potters, Weavers, Blacksmiths)	₹15,000 tool kit incentive + ₹1 Lakh (Tranche 1) & ₹2 Lakh (Tranche 2) @ 5% concessional interest.	1-Click eligibility check, craft certification pre-fill, and application tracking.
Pehchan Artisan ID Card	All verified handicraft & handloom artisans	Official national artisan identity, free accidental insurance coverage (₹2 Lakh), and direct haat stalls.	Integrated directly into the Product Passport QR code as proof of master craft lineage.
Ambedkar Hastshilp Vikas (AHVY)	Artisan Self-Help Groups (SHGs) and rural craft clusters	Common Facility Centers (CFC), raw material bank subsidies, and modern design workshops.	Facilitator/NGO desk clusters artisans into compliant SHGs to claim bulk infrastructure grants.
Pradhan Mantri Mudra Yojana	Micro-enterprises & individual artisans	Shishu (up to ₹50,000) & Kishore (up to ₹5 Lakh) collateral-free working capital loans.	Artisan Productivity Score serves as an alternative non-traditional creditworthiness benchmark.



Section 5: Measurable Social Impact & Market Viability

Artisan Income Multiplier (+140% Real Take-Home)

By eliminating the 4-layer middleman chain (Village broker → City wholesaler → Metro retailer → Export agent), direct-to-consumer provenance sales boost the artisan's take-home earnings from ₹350 to ₹850 on an authentic Bagru dupatta.

Zero Recurring Cloud Costs for NGOs & Self-Help Groups

Because the entire AI pipeline executes locally on consumer-grade gaming hardware (RTX 3050 CUDA) or in-browser WASM, rural cooperatives can operate the platform indefinitely with ₹0 monthly API subscriptions.